

Reflux Sentinel — Companion Web App (PWA)

Reference

Access: Connect phone to Sentinel-AP Wi-Fi · Open `http://192.168.1.1`

Platform: Flask PWA · iOS + Android · Install prompt on first visit

Connectivity: Local hotspot only — no internet, no cloud, no external access

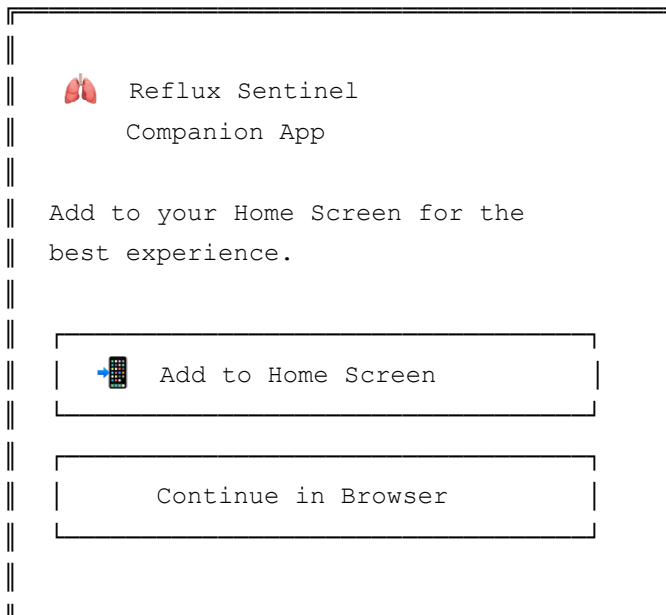
Max connections: 1–2 phones simultaneously

Navigation Structure

Bottom Nav Bar (persistent)

[🏠 Dashboard] [🍴 Food Log] [😴 Sleep] [📊 Reports] [⚙️ Settings] |

1 · First Visit — Install Prompt

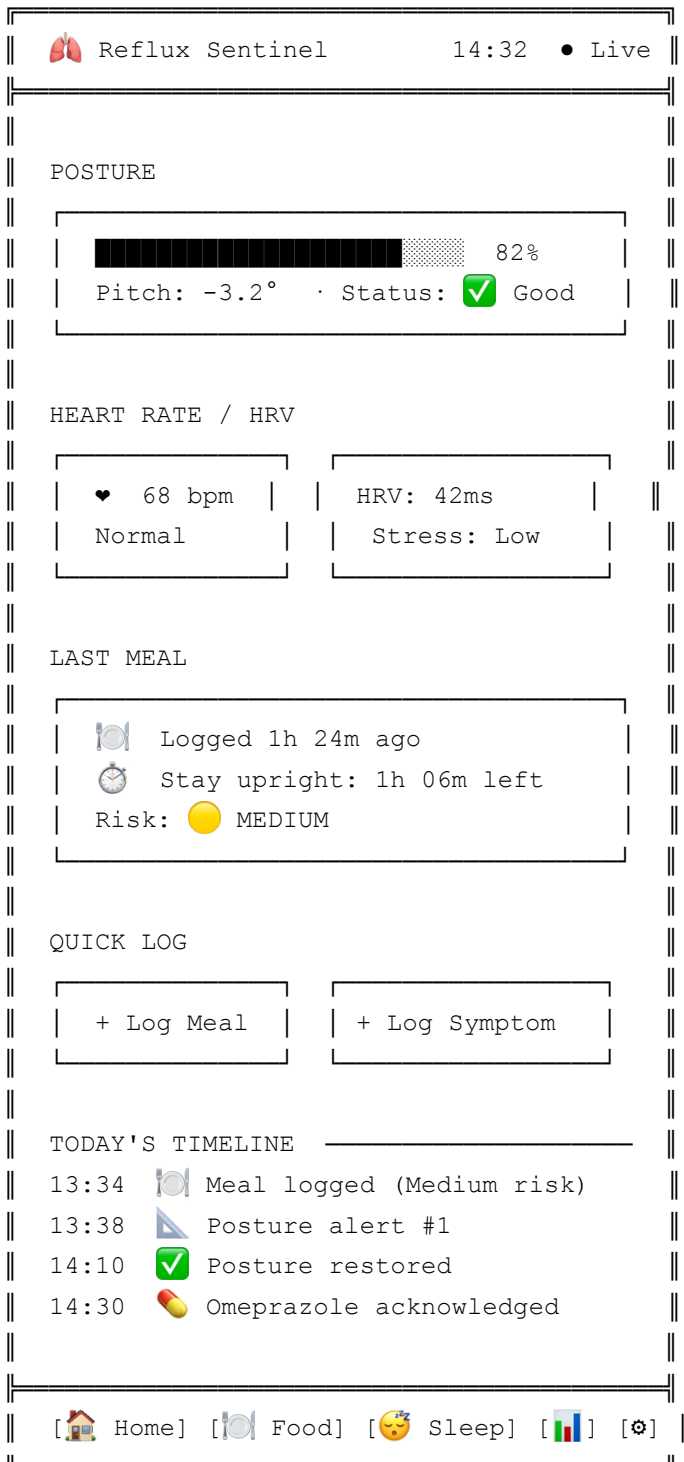


Platform detected: shows native iOS share-sheet instructions or Android "Add to Home

Screen" prompt.

2 · Dashboard (Home)

2.1 Main Dashboard View



2.2 Quick Log — Meal (modal overlay)

+ Log Meal

x

Time: [15:10] (tap to change)

Notes (optional):

e.g. pasta, coffee, large portion

 Attach food photo?

Choose file / Use camera

Save Meal Log

2.3 Quick Log — Symptom (modal)

+ Log Symptom

x

Time: [15:14]

Symptom type:

☒ Heartburn

☐ Regurgitation

☐ Bloating

☐ Chest pain

☐ Other: [_____]

Severity:

☐ 1 Mild

☒ 2 Moderate

☐ 3 Severe

Notes (optional):

Save Symptom Log

3 • Food Log

3.1 Food Log List

 Food Log

Filter: [Today ▼] [All risks ▼]

Sun 5 Apr 2026

13:34  Pizza (detected)
Risk:  HIGH
Upright 2h 58m 
[\[View details\]](#)

08:10  Oats (manual)
Risk:  LOW
Upright 2h 12m 

Sat 4 Apr 2026

19:45  Pasta (detected)
Risk:  MEDIUM
Upright: 1h 10m 

+ Add manual entry

3.2 Food Entry Detail

pasta

MED

2.5h

[+ Add custom entry]

Save Changes

4 · Sleep Chart

4.1 Sleep Overview

🛌

Sleep

Last night: Sat → Sun

Duration detected: 7h 12m

POSITION CHART (all night)

23:00 ————— 06:12

LEFT

RIGHT

BACK

FRONT

SUMMARY

● Left side

6h 01m

● Right side

1h 11m

Haptic nudges sent: 1 / 3 max

Nudge at: 02:18 (right-side detected)

Repositioned within: 4 min

Prev night ◀

▶ Tonight (live)

5 · Reports

5.1 Reports Overview

 Reports

 MORNING REPORT

Sun 5 Apr 2026

Meals: 3 · Symptoms: 1 · Sleep: 

Avg posture: 79% · Alerts: 2

[View Full Report]

 7-DAY TRENDS

Posture avg:  +12% ↑

Symptoms:  -4 ↓ 

Sleep left:  +52% ↑ 

 EXPORT DATA

Export CSV

Export JSON

Generate PDF Report (on device)

Date range: [Last 7 days ▼]



5.2 Morning Report (full view)

◀ Morning Report · Sun 5 Apr	
POSTURE	
Average compliance:	82%
Alerts triggered:	2
Worst period:	13:38-13:51
MEALS	
Total logged:	3
High-risk:	1 (pizza, 13:34)
Upright time maintained:	2/3
SYMPTOMS	
Total:	1 (heartburn, moderate, 15:14)
Correlated to:	pizza meal (+1h 40m)
SLEEP	
Duration:	7h 12m
Left-side:	83%
Nudges sent:	1
HRV / STRESS	
Resting avg:	42ms
Peak stress:	14:10 (+12ms post-meal)
MEDICATIONS	
Omeprazole 20mg:	Taken (20:03)
[Export PDF] [Share]	

6 · Settings

6.1 Settings Root

⚙ Settings
DEVICE

Wear side	[Left ▼]
Calibrate posture	[Run now →]
Hotspot schedule	[07:00-23:00 →]

ALERTS & NUDGES

Posture threshold	[15° —●—]
Alert cooldown	[5 min —●—]
Day haptics	[ON ●]
Voice alerts	[ON ●]

SLEEP

Sleep window	23:00-07:00
Auto-detect sleep	[ON ●]
Night nudges (max)	[3 —●—]
Grace period	[2 min —●—]

MEALS

Upright reminder	[ON ●]
Default window	[2.5h —●—]
Food risk table	[Edit →]

MEDICATIONS

[Manage medications →]

DATA

Auto-backup on connect	[ON ●]
Clear all data	[Delete →]

6.2 Medication Management

◀ Medications

 Omeprazole 20mg
Every day at 20:00
Reminder: ON ☒ [Edit] [Del]

 Gaviscon Advance
Every day at 22:00
Reminder: ON ☒ [Edit] [Del]

+ Add Medication

6.3 Add / Edit Medication (modal)

Add Medication
×

Name:

e.g. Omeprazole 20mg

Dose / Notes (optional):

e.g. Take with water

Time: [20:00]

Frequency:
☒ Daily
☐ Weekdays
☐ Custom

Reminder on OLED: [ON ☒]

Save

6.4 Hotspot Schedule

⚠ **Primary control is on the device** (Settings → Hotspot → Set Schedule on OLED + encoder).

The companion app shows current status read-only, and allows editing schedule hours when already connected.

◀ Hotspot Schedule

Current mode (set on device):

- Scheduled ○ Always on ○ Manual

Schedule hours (edit here or on device):

ON at: [07:00]

OFF at: [23:00]

Save to device

ⓘ Toggle hotspot on/off instantly via the tactile button on the device.

6.5 Calibration Status

⚠ **Calibration is triggered on the device** (Settings → Calibrate on OLED, or hold encoder).

The companion app shows the last saved baseline read-only — useful for confirming a calibration happened.

◀ Calibration

Last baseline (saved on device):

Pitch offset: -1.2°

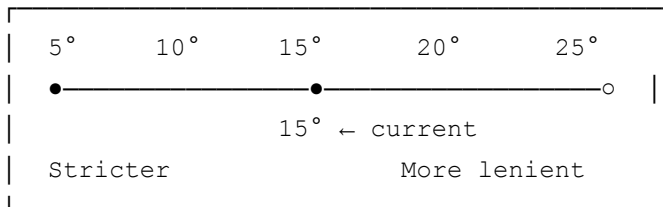
Captured: Sat 4 Apr, 09:15

To recalibrate:

- 1. Stand upright with device clipped on
 - 2. Go to Settings → Calibrate on device
 - 3. Click encoder to capture
- ⓘ Calibration cannot be triggered remotely — physical interaction with the device is required.

7 • Posture Threshold Slider (inline in Settings)

Posture alert threshold



Alert fires when pitch exceeds this angle for more than [30s —●—] continuously

8 • Connection Status (Device-Primary)

Hotspot status is owned by the **device** — the OLED and physical button are the primary controls.

The companion app shows status passively; it cannot reach the device when the hotspot is off.

8.1 OLED — Hotspot Status (on device)

Shown in Settings → Hotspot, and as a brief overlay when toggled.

```
HOTSPOT
-----
Status:  ● ON
192.168.1.1
Connected: 1 phone
▶ Toggle Off
```

```
| Set Schedule |
```

```
| HOTSPOT |
| _____ |
| Status:  ○ OFF |
| Scheduled ON at: 07:00 |
|
| ▶ Turn On Now |
|   Set Schedule |
```

8.2 Companion App — Connection Banner (read-only)

Persistent banner at top of every page. Informational only — no controls.

When connected:

```
|| ● Device online · Last sync: just now ||
```

When hotspot is off / out of range:

```
|| ○ Device offline · Last sync: 2h ago ||
|| Toggle hotspot on the device to connect ||
```

No “Turn on hotspot” button in the app — the physical device button or OLED toggle is the only mechanism.

9 • PWA Page Map

```
http://192.168.1.1/
|
| — /           Dashboard (live posture, HRV, timeline, quick log)
| — /food       Food log (list, filters, entry detail, photo)
| — /sleep      Sleep chart (position all night, summary)
| — /reports    Morning report, 7-day trends, export
| — /settings
```

└─ /settings/device	Wear side, calibrate, hotspot schedule
└─ /settings/alerts	Thresholds, cooldowns, haptic/voice toggles
└─ /settings/sleep	Sleep window, grace period, night nudge max
└─ /settings/meals	Upright window, food risk table editor
└─ /settings/meds	Add / edit / delete medication reminders
└─ /settings/data	Auto-backup toggle, export, clear data

10 • Design Notes

Property	Value
Base URL	<code>http://192.168.1.1</code>
Backend	Flask (Python)
Config reload	watchdog hot-reload of <code>config.json</code>
Platform	iOS + Android via standard browser
Offline mode	Not applicable (device is the server)
Data format	SQLite on device; exports as CSV / JSON
PDF generation	Rendered on phone from JSON payload
Max connections	1–2 simultaneous phones
No external calls	All data stays local; zero cloud

All screens rendered at mobile viewport (~390px wide). Dark mode recommended for legibility. No authentication layer — physical proximity to the hotspot is the security boundary.